1. Consider the following proposition, which we could label as $P(n)$.
   If $5n+3$ is even, then n is odd.

   Prove by strong induction that for every positive integer n, proposition $P(n)$ is true.

   $Proof$:
   By strong induction, assume that $P(k)$ is true for $1 \leqslant k \leqslant n-1$
   The base case of $k=1$ holds as $P(1)=8$ which is even, and 1 is odd.
   As does the base case of $P(2)=13$, which is odd, and 2 is even.
   $5(n-2)+3=(5n+3)-10$
   As $5n+3$ is even, as is 10, $5(n-2)+3$ is therefore even.
   This proves the inductive hypothesis as $P(n-3) \implies P(n-2)$ ■

   I don't love this proof. If someone wants to touch it up, they are welcome to.

2. Prove by strong induction that for every positive integer n the following holds:

$\sum\limits_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$

proof:

By strong induction, assume:

$\sum\limits_{k=1}^{n-1} k(k+1) = \frac{n(n+1)(n-1)}{3}$

The base case $n = 1$ holds:

$\sum\limits_{k=1}^{1} k(k+1) = 1(1+1) = 2 = \frac{1(1+1)(1+2)}{3}$

By subsituiting the last term of the summation:

$$\sum_{k=1}^{n} k(k+1) = \frac{n(n-1)(n+1)}{3} + n(n+1)$$

The inductive hypothesis is proved below:

$$\frac{n(n-1)(n+1)}{3} + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

$n(n-1)(n+1) + 3n(n+1) = n(n+1)(n+2)$

$n^3 - n + 3n^2 + 3n = n^3 + 3n + 2n$

$n^3 + 3n^2 + 2n = n^3 + 3n^2 + 2n$

■